

that can deploy an open source digital government framework (OSDGF). We will also explore appropriate ways to map open source technologies to the various architecture layers.

Strategic concerns of OSS in e-governance

Figure 1 depicts the drivers, services from stakeholders and the architecture in the context of the strategic considerations of OSS. The drivers are flexibility, cost savings, security, freedom to reuse, etc, which contribute to a government's economic growth.

Flexibility: Government agencies can tailor open source products, adding to or modifying the code to fit their specific needs. These modifications can include security patching for critical vulnerabilities, which can be completed in hours.

Savings: Open source software is usually available at a much lower price point than commercial products. And valuable support options are available for popular packages.

Freedom to use and reuse: Open source licences do not limit or restrict who can use the software, the type of user, or the areas of business in which the software is used.

Security: It is very critical to implement proper security features in e-governance initiatives to eliminate and minimise the fraud in various schemes and policies that will be implemented across the country. Open source software can be used to implement security quickly and more transparently.

Innovation: The cloud, Big Data, cyber security and API management are seeing a much faster cycle of churn with respect to technology choices for the next generation of government initiatives. And it is open source that is providing most of the required capabilities, much faster and cheaper than commercial vendors.

High quality: Support from a good developer community assures the government of high quality software.

Skills development: Using OSS allows the government to unlock a wider base of prospective IT talent in the open source space.

Classification of the stakeholders

Listed below are the key stakeholders in the open source domain:

1. Collaborators (different communities of the government)
2. Consumers and contributors (citizens, government

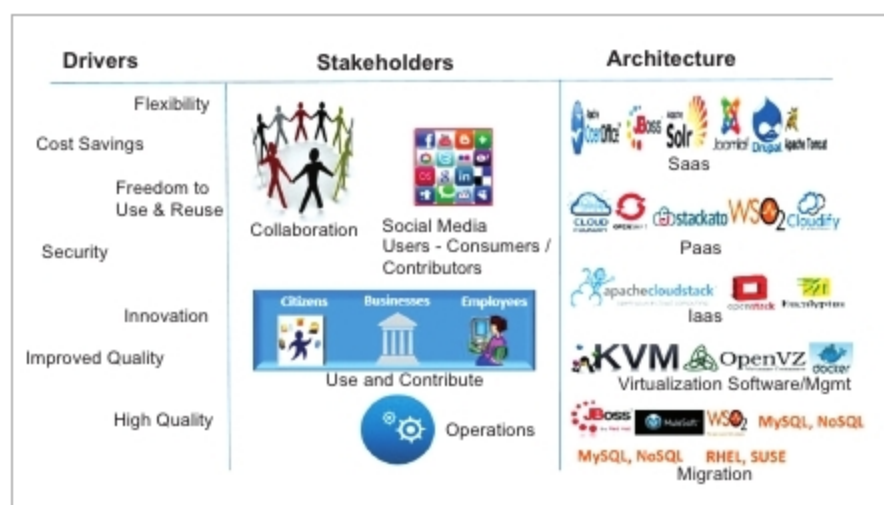


Figure 1: Examples of the strategic considerations of OSS

servants and social media

3. Users and contributors (developers, citizens, businesses, and industry)
4. Users and maintainers (operations)

Opportunities for the use of OSS lie in Software as a Service, Platform as a Service and Infrastructure as a Service.

The characteristics of OSS that make it ideal for e-governance

With OSS, governments can avoid vendor lock-in and ensure seamless integration of disparate applications, products and systems that have been developed and deployed by different organisations and vendors.

The source code is available to the community, the adapters and the end user to study and modify the software and to redistribute copies of either the original or the modified versions. In the case of OSS, the source code is free from any royalty.

There is no vendor monopoly when using free and open standards. With data transferability and open data formats, there are greater opportunities to share data across interoperable platforms. The adoption of OSS enhances interoperability with other e-governance solutions because of the reuse of recommended software stacks, libraries and components.

Figure 2 depicts the high-level characteristics of OSS.



Figure 2: The characteristics of OSS

The philosophy behind adopting OSS for digital governance

OSS based e-governance architecture should create an environment that provides the following:

- A self-healing application architecture that ensures an automated response in the event of a failure or a traffic surge. This is due to the business process manager, which automates actions with minimal or no manual intervention.
- A microservices based horizontal scaling approach, which creates a larger application functionality than a monolithic application block. This ensures that only parts of the application to be scaled can get additional capacity for growth.
- Standardised platforms to rationalise skills and technologies. This ensures that there is uniformity in approach and readily available skillsets without